Geometry, Entropy, and Operator Learning with Tunable Kernels

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Abstract

We study a two-dial family of kernels

$$k_{\theta,\sigma}(y \mid x) \propto w_\theta\big(\|x-y\|\big)\; g_\sigma(y-x),$$

whose product form covers any radial weight w_θ and centred residual density g_σ that satisfy minimal smoothness and moment conditions. To give the theory a concrete footing—and to obtain closed-form bias constants in our finite-sample analysis—we instantiate this family with a data-adaptive *local-linear* construction, but all results hold for the broader class.

For every observable f we compute the one-step cumulant-generating function (CGF) and show that its quadratic coefficient $\mathcal{I}_{\theta,\sigma}[f]$ serves simultaneously as (i) a *Fisher load*, (ii) $\frac{2}{\theta^2}$ times the synthetic Ricci curvature $\Gamma_2^{(\theta,\sigma)}(f)$, and (iii) $-2\,\partial_\sigma$ of the one-step Shannon entropy. This *Fisher–curvature–entropy bridge* yields a single quadratic objective whose minimiser in the dial plane, $(\theta^\star, \sigma^\star)$—the *little-rose*—optimises curvature bias, entropy production, and operator-learning risk at once. We prove

$$\nabla_{(\theta,\sigma)}\mathrm{KL}\big(\mu_1 \,\|\, k_{\theta,\sigma}\#\mu_0\big) \;=\; -\tfrac{1}{2}\,\nabla_{(\theta,\sigma)}\mathcal{I}_{\theta,\sigma},$$

so steepest KL descent converges to the little-rose with no hyper-parameters. A data-driven Goldenshluger–Lepski selector recovers $(\hat{\theta}_\star, \hat{\sigma}_\star)$ from finite samples, and we give non-asymptotic risk bounds for both Perron–Frobenius and Koopman estimates.

The resulting dial matrices $(\Theta_{ij}, \Sigma_{ij})$ form a directed, doubly-weighted graph, opening avenues toward discrete curvature flows, entropy production on cycles, and multiparameter persistent homology—pointing to a broad research programme at the interface of statistical dynamics, information geometry, and topological data analysis.

1 Introduction

1.1 Why tune local kernels?

1.2 Key contributions

- Closed-form bridge: CGF curvature → Fisher load → curvature & entropy.
- Single KL optimisation (little-rose) jointly tunes **two** dials (θ, σ).
- Finite-sample Goldenshluger–Lepski selector with risk guarantees.

1.3 Organisation of the paper

2 Rose kernels and dials

2.1 Product-form kernels and minimal assumptions (K0–K3)

2.2 Local–linear residual construction (canonical example)

2.3 Scaled kernels and notation for the rest of the paper

3 Population theory

(K0)–(K2) as in Sec. 3.0 — Markov, score L^2, second-moment scaling.

3.1 CGF $\Rightarrow$ Fisher $\Rightarrow$ Landau

3.2 Fisher–curvature–entropy bridge

3.3 Introducing the dial plane

(K3) C^1 regularity in (θ, σ)